From MIMIC-IV to a guideline-adherence judgment: cohort flow, rule windows, and the leak-free case card

A Cohort construction (MIMIC-IV v3.1, numbers = admissions)

MIMIC-IV v3.1 (hosp + icu modules)
364,627 patients · 546,028 admissions · 94,458 ICU stays

First ICU stay per patient · adults ≥ 18 y
65,366 patients, one admission each (no readmissions)

Diagnosis codes, any position (ICD-9 / ICD-10)

Pneumonia 8,234	UTI 7,793
ICD-10: J13–J16, J18.0/.1/.8/.9	ICD-10: N10, N12, N39.0
ICD-9: 481–483, 485, 486	ICD-9: 590.1/.80/.9, 599.0

Excluded
aspiration 815 · VAP 419 (pneumonia)
catheter-related 330 (UTI)
both pneumonia and UTI codes 1,029

After exclusions
Pneumonia 6,056 · UTI 6,434

Suspicion of infection = culture ↔ antibiotic pairing
culture_first (culture → antibiotic ≤ 72 h): 4,752 · 5,464
antibiotic_first (antibiotic → culture ≤ 24 h): 1,009 · 628
with a suspicion time: 5,761 · 6,092

no pairing 295 · 342

Community-acquired: suspicion ≤ 48 h after hospital arrival
Pneumonia 4,875 · UTI 4,523 (final cohort)

hospital-acquired, > 48 h: 886 · 1,569

Sepsis stratification — three labels (pneumonia · UTI)

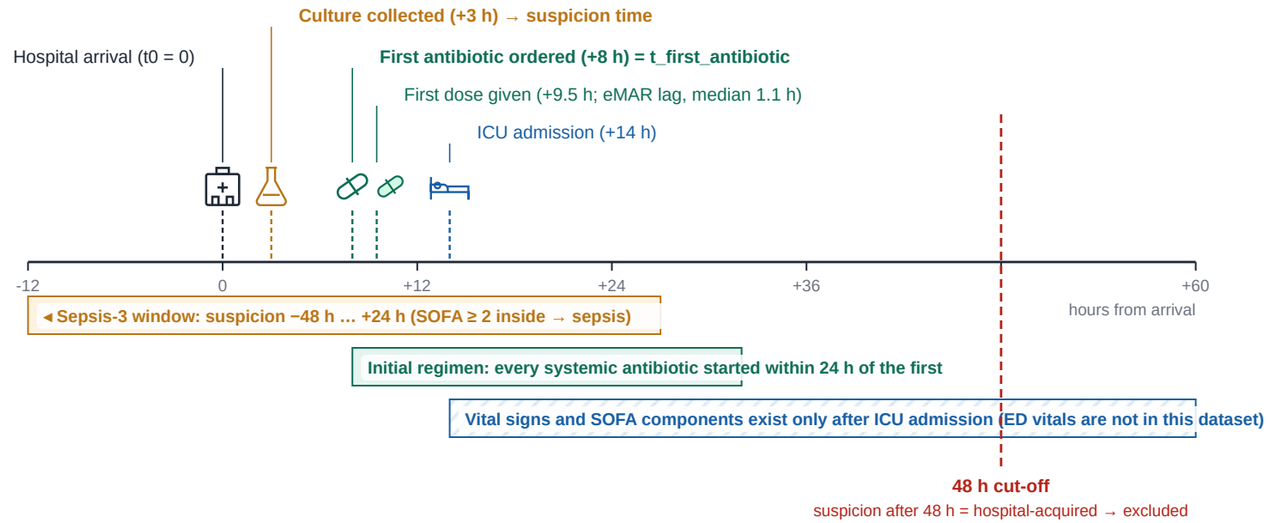
sepsis3 (mimic-code, primary)	**3,666 · 3,180**
simplified SOFA ≥ 2 (shared code, also for K-MIMIC)	**3,858 · 3,496**
ICD sepsis codes	**1,721 · 1,526**

agreement with sepsis3: SOFA 85% · 82% / codes 50% · 54%
18% have no ICU data inside the SOFA window → flagged, not excluded

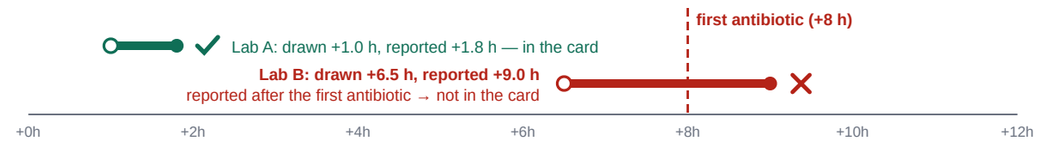
Case cards → LLM agents (panel C)
20 evaluation cases (5 × 4 groups) + 20 backup · hashed IDs · relative times

B One patient's timeline and the rule windows (hours from hospital arrival, t0 = 0)

Rule 1 · culture_first culture ≤ 72 h antibiotic ⇒ **suspicion time = culture time**
Rule 2 · antibiotic_first antibiotic ≤ 24 h culture ⇒ **suspicion time = antibiotic time**



Availability rule for the case card: a value enters only if it was reported (storetime) before the first antibiotic



If chart time were used instead: 13% of 'last values' and 21% of patients would carry results the clinician had not seen

C Case card for the agent (hashed ID, all times relative to the first antibiotic = T+0)

case.json

- Age, sex, weight · ED visit · ICU type
- Vitals and labs: last value reported before T+0
- Cultures before T+0: specimen and time only, no results
- Empirical set: ingredient · class · AWaRe · route · dose · start T+h
- History: admission / IV antibiotics ≤ 90 d · MRSA / Pseudomonas ≤ 1 y
- Comorbidities from codes (DM, CKD, COPD, immunosuppression, …)

Not in the card

- ✗ Discharge diagnoses
- ✗ In-hospital death
- ✗ Culture results after T+0
- ✗ SOFA score / sepsis label

Judgment
A1 syndrome & severity → A4 adherence vs KR / INTL guideline → A5 grounding check
Clinicians rate 5 cases × 4 groups (1–5 and 4-level verdict)